The effect of the use of artificial intelligence by managers on the operational indicators of IT companies: analysis of the largest Russian firms.

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Research Motivation

- AI drives digital transformation, improves operational performance, and enhances competitive performance via productivity gains, innovation, and actionable insights. **(Mikalef & Gupta, 2021; Enholm et al., 2022; Cannas et al., 2024; Mikalef et al., 2020; Babina et al., 2024)**
- AI benefits grow with complementary digital capabilities (analytics, infrastructure, knowledge). **(Abrokwah-Larbi & Awuku-Larbi, 2024; Lee et al., 2019; Lin & Sun, 2026)**
- AI's impact depends on integration into managerial decision-making and operations – not just adoption. **(Wang et al., 2025)**
- Most studies examine general AI adoption in developed economies; the role of managers in shaping operational indicators remains underexplored – particularly in the Russian IT sector, a dynamic, sanctions-exposed, and empirically overlooked context. **(Enholm et al., 2022; Mikalef & Gupta, 2021; Molodchik et al., 2024)**
- Firm size may shape AI-performance links: larger firms possess greater data volumes, digital infrastructure, and financial resources to convert managerial AI use into operational outcomes – a moderating effect this study explicitly tests. **(Ardito et al., 2024; Lin & Sun, 2026; Molodchik et al., 2024)**
- Prior studies use longer time windows (>10 years); they miss 2019–2025: COVID-19, sanctions, Russian IT reconstruction. **(Molodchik et al., 2024; literature gap)**
- This study tests whether positive AI-performance links hold under distinct institutional/temporal conditions and whether firm size moderates the relationship.

Research Questions

(1)

How does the use of artificial intelligence by managers relate to the operational indicators of large Russian IT companies?

(2)

How does firm size moderate the relationship between managerial AI integration and the operational indicators of large Russian IT companies?

Theoretical Framework

Resource-Based View (RBV)

Firm performance depends on valuable and unique resources.

Artificial intelligence can be viewed as a strategic organizational capability that improves firms' operational performance when combined with complementary resources.

(Wernerfelt, 1984; Barney, 1991)

Dynamic Capabilities Theory

Firms improve performance by reconfiguring resources and adapting to changing environments.

AI technologies help managers enhance decision-making, improve resource allocation, and optimize operational processes.

(Teece, Pisano & Shuen, 1997)

Contingency Theory

There is no one best way to organise; organisational effectiveness depends on the alignment between internal structures and contextual conditions.

Managerial AI integration does not guarantee performance improvements – its effect on operational indicators is moderated by firm size, which shapes the firm's ability to convert AI insights into outcomes.

(Woodward, 1965)

Hypotheses Development

Hypothesis 1

• AI technologies enable firms to process large data, automate operations, and support managerial decision-making with predictive insights and real-time analysis. **(Mikalef & Gupta, 2021; Mikalef et al., 2020; Wamba-Taguimdje et al., 2020)**

• AI adoption contributes to higher productivity, operational efficiency, and better firm performance by optimising resource allocation through data-driven management practices. **(Babina et al., 2024; Zhou et al., 2026; Pastor-Merino et al., 2026; Mikalef & Gupta, 2021)**

• Most evidence on AI-performance links comes from developed economies; the Russian IT sector remains unexplored, so it is unclear whether positive effects hold in this institutional environment. **(Molodchik et al., 2024)**

• Large Russian IT companies face sanctions, market restructuring, and rapid digital transformation – unique conditions that may alter the effect of managerial AI use on operational indicators. **(Molodchik et al., 2024)**

H1. The use of artificial intelligence by managers positively relates to the operational indicators of large Russian IT companies.

Hypotheses Development

Hypothesis 2

• Larger firms have greater access to data, financial resources, and scale advantages, enabling them to derive stronger benefits from AI investments. **(Mikalef & Gupta, 2021; Babina et al., 2024)**

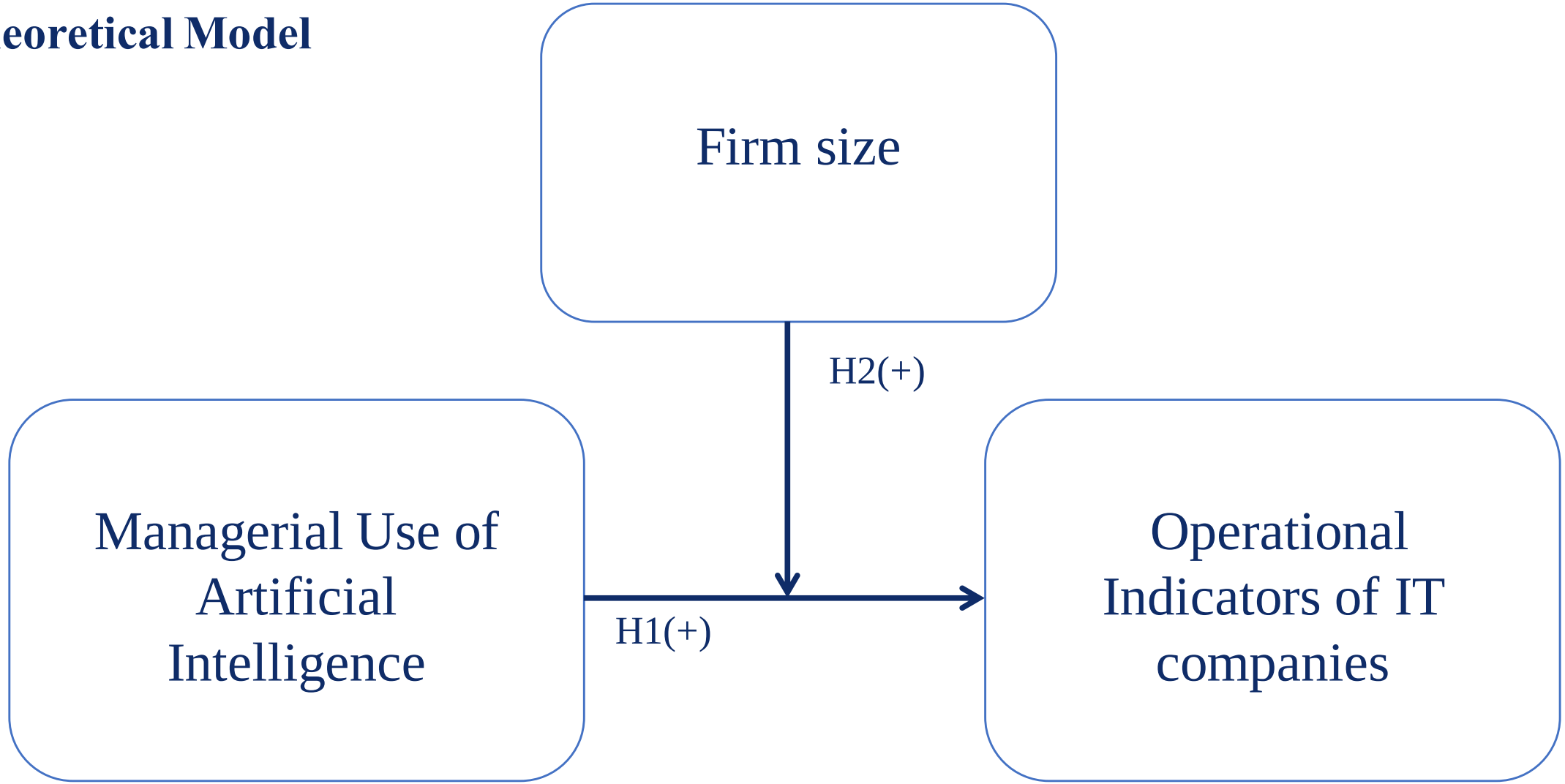
• AI produces more pronounced performance effects in large enterprises than in SMEs, and this positive relationship becomes stronger as firm size increases. **(Babina et al., 2024; Lin & Sun, 2026)**

• In Russian companies, ICT adoption (including AI) shows stronger financial effects in technology-intensive sectors like IT, where larger firm size correlates with greater ICT complexity. **(Molodchik et al., 2024)**

• Large Russian IT firms face sanctions and market restructuring; their scale provides resilience and resource capacity, making size a critical moderating factor. **(Molodchik et al., 2024)**

H2. Firm size positively moderates the relationship between the use of artificial intelligence by managers and operational indicators of large Russian IT companies, so that this relationship is stronger for larger firms.

Theoretical Model



Data, Sample and Measures

- **Research design:** quantitative study based on panel data from large Russian IT companies
- **Data sources:** annual reports, financial statements, official websites, and company press releases
- **Period of observation:** 2019–2025
- **Final sample:** 23 large Russian IT companies
- **Dataset structure:** 161 firm-year observations and 20 variables
- **Sample selection:** non-random, criterion-based sample selected by criteria: IT sector relevance, large market scale, and availability of comparable public data for 2019–2025. Firm size was assessed using : revenue-based lists, and the latest-year revenue range in the final sample: from RUB 5 billion.
- **Limitation:** the findings cannot be generalized to the entire Russian IT sector due to limited disclosure among private companies

Measures

	Variable	Formula	Unit of measurement	Data source
IV	Managerial AI Integration	AI manager mentions / Total AI mentions	Unitless ratio, ranging from 0 to 1	Annual reports, official websites, corporate press releases
DV: Operational Indicators	Labor Productivity	Revenue / Number of employees; then log-transformed for regression analysis	Million RUB per employee; log-transformed	Financial statements, annual reports
	ROA	Net profit / Total assets	Unitless financial ratio	Financial statements
	Capital Investment Intensity	Capital expenditures / Total assets	Unitless financial ratio	Financial statements, annual reports
Moderator	Firm Size	Total assets; then log-transformed for regression analysis	Million RUB of total assets; log-transformed	Financial statements
Control variable	Firm Age	Observation year − Foundation year	Number of years	Company websites, corporate profiles, public company information
	Leverage	Total liabilities / Total assets	Unitless financial ratio	Financial statements
	Year Effects	Year dummies for 2020–2025; 2019 is the base year. Values: 1 = observation year, 0 = otherwise.	Binary variables: 0 or 1	Constructed from the panel dataset

Analysis

- **R-studio** software
- **For preliminary analysis:** Data cleaning, Descriptive statistics, Correlation matrix
- **For main analysis:** Regression models with robust standard errors and year effects

Descriptive statistics and correlation matrix

Variable	Min	Max	Mean	SD	1	2	3	4	5	6
1. Labor productivity	-0.33	9.39	2.32	1.14	1					
2. ROA	-0.41	3.36	0.18	0.38	-0.01	1				
3. Capex intensity	0.01	0.80	0.15	0.13	0.19**	-0.14*	1			
4. Managerial AI integration	0	1	0.19	0.29	-0.08	-0.23***	0.05	1		
5. Firm size	5.01	18.05	11.56	3.01	0.21**	-0.5***	0.28***	0.41***	1	
6. Firm age	4	34	22.33	7.85	-0.1	-0.32***	-0.04	0.06	0.26***	1

*Note. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.*

Hypothesis 1

"Managerial use of artificial intelligence is positively related to the operational indicators of large Russian IT companies."

DV: labor productivity, ROA, capex intensity
IV: managerial_ai_integration
Controls: firm size, firm age, leverage, year effects.
Sample: panel of 23 largest Russian IT companies, 2019–2025; 149 obs after outlier removal

Main analysis Hypothesis 1

Variable	Model1	Model2	Model3
Main effects			
Managerial AI integration	-0.91** (0.327)	-0.09 (0.195)	-0.08 (0.084)
Controls			
Firm size (log assets)	0.18*** (0.032)	-0.06*** (0.018)	0.02 (0.008)
Firm age	-0.03*** (0.011)	-0.01 (0.006)	-0.01** (0.003)
Leverage	0.18 (0.353)	-0.24 (0.188)	0.04 (0.081)
Year effects	Yes	Yes	Yes
Model fit			
Adjusted R^2	0.21	0.11	0.03
F(df)	4.89** (10, 138)	2.92** (10, 138)	1.53 (10, 138)

*Note. Unstandardized coefficients; standard errors in parentheses.*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.*

The dependent variables are operational indicators of large Russian IT companies: log labor productivity, ROA, and capital investment intensity.

The independent variable is managerial AI integration.

Control variables include: firm size, firm age, leverage, and year effects.

Model 1: Log labor productivity.

Model 2: ROA.

Model 3: Capital investment intensity.

VIF diagnostics indicate no multicollinearity concerns, as all values are below 2, with the maximum VIF equal to 1.31.

Year effects are included in all models.

Hypothesis 2

"Firm size positively moderates the relationship between managerial use of AI and operational indicators, so that the relationship is stronger for larger firms."

IV: managerial_ai_integration
Moderator: firm size
Interaction term: managerial_ai_integration × log_assets
DV: labor productivity, ROA, capex intensity
Sample: panel of 23 largest Russian IT companies, 2019–2025; 149 obs after outlier removal

Main analysis Hypothesis 2

Variable	Model 1	Model 2	Model 3
Main effects			
Managerial AI integration	-0.264	-0.018	-0.01(0.035)
Firm size(log assets)	0.18*** (0.053)	-0.05*** (0.012)	0.02*** (0.004)
Interaction			
Managerial AI integration×Firmsize	-0.12(0.119)	0.11** (0.039)	-0.03(0.014)
Controls			
Firmage	-0.00045	-0.000055	-0.002(0.001)
Leverage	0.10 (0.262)	-0.11(0.091)	-0.0033
Year effects	Yes	Yes	Yes
Constant	2.66*** (0.406)	0.29** (0.112)	0.29*** (0.052)
Model fit			
Adjusted R^2	0.21	0.33	0.09
F(df)	4.55*** (11,137)	7.58*** (11,137)	2.31* (11,137)

*Note. Unstandardized coefficients; standard errors in parentheses.*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.*

The dependent variables are operational indicators of large Russian IT companies: log labor productivity, ROA, and capital investment intensity.

The independent variable is managerial AI integration.

The moderator is firm size, measured as the logarithm of total assets.

The interaction term is Managerial AI integration × Firm size.

Managerial AI integration and firm size were mean-centered prior to computing the interaction term.

Model 1: log labor productivity.

Model 2: ROA.

Model 3: capital investment intensity.

Control variables include firm age, leverage, and year effects.

VIF diagnostics indicate no multicollinearity concerns, as all values are below 2, with the maximum VIF equal to 1.44.

Results: Summary

Hypothesis		Coefficient (SE)	Outcome
H1	Managerial AI integration → operational indicators	Labor productivity: -0.91**(0.33)	Not supported
		ROA: -0.10 (0.20)	
		Capex intensity: -0.08 (0.08)	
H2	Managerial AI integration × Firm size → operational indicators	Labor productivity: -0.12 (0.12)	Partially supported
		ROA: 0.11** (0.04)	
		Capex intensity: -0.03 (0.014)	

For H1: Adjusted R^2 = 0.21 / 0.11 / 0.03; F(df) = 4.89** / 2.92** / 1.53. For H2: Adjusted R^2 = 0.21 / 0.33 / 0.09; F(df) = 4.55*** / 7.58*** / 2.31*. *** p < 0.001; ** p < 0.01; * p < 0.05.

Managerial AI integration does not show a consistent positive effect on operational indicators. Firm size partially strengthens this relationship, but only for ROA.

Discussion

Why H1 was not confirmed

1. Managerial AI integration did not show a positive significant effect on operational indicators.
2. AI use by managers may not immediately translate into measurable improvements in labor productivity, ROA, or investment activity.
3. The negative result for labor productivity may reflect implementation costs, adaptation delays, and the need for organizational restructuring.
4. Public AI-related disclosures may capture the declared presence of AI rather than its deep integration into daily managerial decision-making.
5. AI creates value only when it is supported by complementary resources, data infrastructure, employee capabilities, and established managerial routines.

Why H2 was partly confirmed

1. Firm size strengthened the relationship between managerial AI integration and ROA, but not with labor productivity or capital expenditure intensity.
2. Larger firms may have more financial, technological, and human resources to convert AI-supported decisions into profitability-related outcomes.
3. The absence of significant moderation for labor productivity and capex intensity shows that firm size does not improve all operational outcomes equally.
4. The moderating role of firm size is outcome-specific: it matters for profitability, but not for productivity or investment intensity.
5. This supports the idea that the value of managerial AI integration depends on organizational context rather than AI use alone.

Discussion

Theoretical Contribution

The study shifts the focus from general AI adoption to managerial AI integration, showing that AI use in managerial decision-making does not automatically translate into better operational performance.

(Mikalef & Gupta, 2021)

The results refine the resource-based view and dynamic capabilities theory: AI creates value only when it is supported by complementary resources, organizational routines, and implementation depth.

(Mikalef & Gupta, 2021; Mikalef et al., 2020)

The study also contributes to contingency theory by showing that firm size matters selectively. It strengthens the relationship between managerial AI integration and ROA, but not labor productivity or capital expenditure intensity.

(Molodchik et al., 2024)

Practical Implications

Managers should not expect AI integration to automatically improve productivity, profitability, or investment activity in the short term. This is consistent with the study results, as H1 was not supported.

AI-related managerial practices should be embedded into concrete functions: planning, forecasting, monitoring, resource allocation, and operational control.

(Mikalef & Gupta, 2021)

Larger firms may be better positioned to convert managerial AI integration into profitability-related outcomes, but size alone is not enough. AI-generated insights must be connected to actual managerial decisions and operational processes. This is consistent with the study results, as H2 was only partially supported.

(Molodchik et al., 2024)

Limitations and further research agenda

Limitation	Further Research Agenda
AI was measured through public disclosures, which may not show actual use.	Use surveys, interviews, or web data for a more accurate measure.
The sample is limited to large Russian IT firms.	Test other firm sizes, industries, and countries.
The study shows associations, not causality.	Use stronger causal designs and longer panel data.
The period includes COVID-19 and sanctions.	Study more stable periods or compare with other contexts.
Mediating mechanisms were not tested.	Examine mediators like decision quality and innovation.